BRAD ZIYI WANG

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Education & Skills

McMaster University Hamilton, ON, Canada
Bachelor of Mechatronics Engineering (Co-op) & Management Est. Graduation: May 2028
• **Cumulative GPA:** 3.70/4.00,
• **Accolades:** Dean's Honors List 2022-2024.
• **Relevant Course Topics:** Electronics and Instrumentation, Digital Communication, Analog and Digital Circuits, Embedded Systems, Data Structures and Algorithms, Control Systems, Signals and Systems, Operating Systems.

Technical Skills

Developing: DevOps, SolidWorks, KiCAD PCB Design, Autodesk Inventor, Git, GitHub, Linux, Microsoft Office, Excel, Photoshop
Manufacturing: Soldering & Crimping, 3D Printing, Technical Drawing, Welding, Laser Cutting, Hand & Power Tools
Coding: C, C++, Verilog, MATLAB, Simulink, Python, R, Julia, LaTeX

Experience

Hardware Engineering Intern, IPG Photonics (Ontario, Canada) May 2026 – Present

• Developing a multi-sensor machine learning system for quality control in laser processing, addressing a gap that existing monitoring methods leave unfilled.
• Designing and building a calibration station for precision optomechanical systems.
• Setting up a new robotic arm lab, from planning through to a working workspace.

Team lead, McMaster Medical Engineering Design Team (Ontario, Canada) September 2024 – May 2026

• Led a 12-member multidisciplinary team (2024-25) to design and prototype a baby safety monitoring system utilizing **Computer Vision and facial recognition algorithms**, resulting in a successful industry-sponsored year-end showcase.
• **Recruited** and led a 12-member team (2025-26) to design and prototype a mechanically switching tip guide cane that performs on snow, sand and other terrain where standard canes struggle, earning 1st place at the year-end showcase.
• **Managed milestones, design reviews, and testing**. Planned and tracked project schedules using **Gantt charts**, ensuring timely progress and alignment with deliverables.

Quality Control Intern, Caterpillar (Jiangsu, China) May 2024 – August 2024

• Led a new **weather-protection protocols** for outdoor storage, preventing corrosion and reducing storm-related damage.
• Leveraged hands-on operational expertise to train technicians on Hexagon laser scanners, **reducing calibration time by 83%** (30 to 5 minutes) and improving equipment readiness.
• Performed **calibration and detailed logging** of 1,000+ torque wrenches and pneumatic/electrical tools in compliance with company and industry standards for seasonal torque checks.
• Collaborated with cross-functional engineering teams in CAT India and CAT USA to **standardize assembly protocols**, reducing seal damage and strengthening global quality compliance through rigorous shop floor audits.

Selected Side Projects

Cybernetic Robot with Q-Learning Machine Learning | C++, STM32, Circuitry, MATLAB July 2025
• Developed a wheeled **reinforcement-learning** robot with a lightweight Q-learning algorithm optimized for limited-memory **STM32** hardware. Designed and simulated closed-loop control systems in **MATLAB Simulink**, integrating real-time sensors.

Laser Range Finder | Arduino IDE, Technical Drawings, Circuitry June 2023
• Created an **Arduino**-based laser range finder using a VL53L0X time-of-flight (ToF) **laser distance sensor**. Implemented **I2C communication protocol** to interface the microcontroller with the peripherals.

First Aid Cryo-Container (MALIC System) | Mechanical Design, Thermal Systems, CAD June 2024

• Engineered an **adaptive temperature-controlled system** for preserving severed limbs of varying sizes, inspired by a real industrial accident. **Modular refrigeration** units maintain optimal temperature of ~4°C until surgical reattachment. Refined prototypes through **iterative design**, resulting in a functional final model.